

Exercise 1

Perform following Programs, write & Run the codes and analyze output. Make separate file for each program.

1. Develop following program and run

Note: enter quit for exiting and end.

```
from nltk.chat.util import Chat, reflections
my_dummy_reflections= {
    "mr"    : "Doctor",
    "hello"  : "hey there",
    "play"   : "played"
}

pairs = [
    [
        r"my name is (.*)",
        ["Hello %1, How are you today ?",""]
    ],
    [
        r"quit",
        ["BBye take care. See you soon :) ","It was nice talking to you. See you soon :)"]
    ],
]

def chatty():
    print("Hi, I'm Chatty and I chat alot ;)\nPlease type lowercase English language to start a conversation. Type quit to leave ") #default message at the start
    chat = Chat(pairs, my_dummy_reflections)
    chat.converse()
if __name__ == "__main__":

    chatty()
```

2. Sample output of program 1

```
Python 3.12.4 | packaged by Anaconda, Inc. | (main, Jun 18 2024,
15:03:56) [MSC v.1929 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.
```

```
IPython 8.25.0 -- An enhanced Interactive Python.
```

```
In [1]: runfile('C:/Users/admin/Desktop/
ArtificialIntelligence_2025-2026/chatbot1.py', wdir='C:/Users/admin/
Desktop/ArtificialIntelligence_2025-2026')
Hi, I'm Chatty and I chat alot ;)
Please type lowercase English language to start a conversation. Type
quit to leave
>my name is mr kshitij
Hello Doctor kshitij, How are you today ?
>
```

3. Run the following chatbot and analyse:

Note: enter quit for exiting and end.

```
from nltk.chat.util import Chat, reflections
```

```
pairs = [
    [
        r"my name is (.*)",
        ["Hello %1, How are you today ?"],
    ],
    [
        r"what is your name ?",
        ["My name is Chatty and I'm a chatbot ?"],
    ],
    [
        r"how are you ?",
        ["I'm doing good\nHow about You ?"],
    ],
    [
        r"sorry (.*)",
        ["Its alright", "Its OK, never mind"],
    ],
    [
        r"i'm (.*) doing good",
        ["Nice to hear that", "Alright :)"],
    ]
]
```

```
],
[
    r"hi|hey|hello",
    ["Hello", "Hey there",]
],
[
    r"(.*) age?",
    ["I'm a computer program dude\nSeriously you are asking me
this?",[

],
[
    r"what (.*) want ?",
    ["Make me an offer I can't refuse",]

],
[
    r"(.*) created ?",
    ["Nagesh created me using Python's NLTK library ","top secret ;)",]
],
[
    r"(.*) (location|city) ?",
    ['Vadodara, Gujarat',]
],
[
    r"how is weather in (.*)?",
    ["Weather in %1 is awesome like always","Too hot man here in
%1","Too cold man here in %1","Never even heard about %1"]
],
[
    r"i work in (.*)?",
    ["%1 is an Amazing company, I have heard about it. But they are in
huge loss these days.",]
```

```

    ],
    [
        r"(.*)raining in (.*)",
        ["No rain since last week here in %2","Damn its raining too much
here in %2"]
    ],
    [
        r"how (.*) health(.*)",
        ["I'm a computer program, so I'm always healthy ",]
    ],
    [
        r"(.*) (sports|game) ?",
        ["I'm a very big fan of Football",]
    ],
    [
        r"who (.*) sportsperson ?",
        ["Messy","Ronaldo","Roony"]
    ],
    [
        r"who (.*) (moviestar|actor)?",
        ["Brad Pitt"]
    ],
    [
        r"quit",
        ["BBye take care. See you soon :) ","It was nice talking to you. See
you soon :)"]
    ],
    ]
def chatty():
    print("Hi, I'm Chatty and I chat alot ;)\nPlease type lowercase
English language to start a conversation. Type quit to leave ") #default
message at the start
    chat = Chat(pairs, reflections)

```

```
chat.converse()
if __name__ == "__main__":

    chatty()
```

4. Run the following chatbot and analyse

```
from nltk.chat.util import Chat, reflections
my_dummy_reflections= {
    "mr" : "Doctor",
    "hello" : "hey there",
    "play" : "played"
}
pairs = [
    [
        r"my name is (.*)",
        ["Hello %1, How are you today ?",]
    ],
    [
        r"Have you (.*)",
        ["I never %1",]
    ],
    [
        r"quit",
        ["BBye take care. See you soon :) ", "It was nice talking to you. See you soon :)"]
    ]
]
def chatty():
    print("Hi, I'm Chatty and I chat alot ;)\nPlease type lowercase English language to start a conversation. Type quit to leave ") #default message at the start
    chat = Chat(pairs, my_dummy_reflections)
```

```
chat.converse()  
if __name__ == "__main__":
```

```
chatty()
```

example : my name is mr mahendra dhoni
hello Dr. Dhoni how are you today
have you play football
I never played football

- 5.
- Develop chatbot for a ecommerce website (having atleast 20 ques and ans).

6. Develop chatbots for website of educational institution having atleast 20 queries